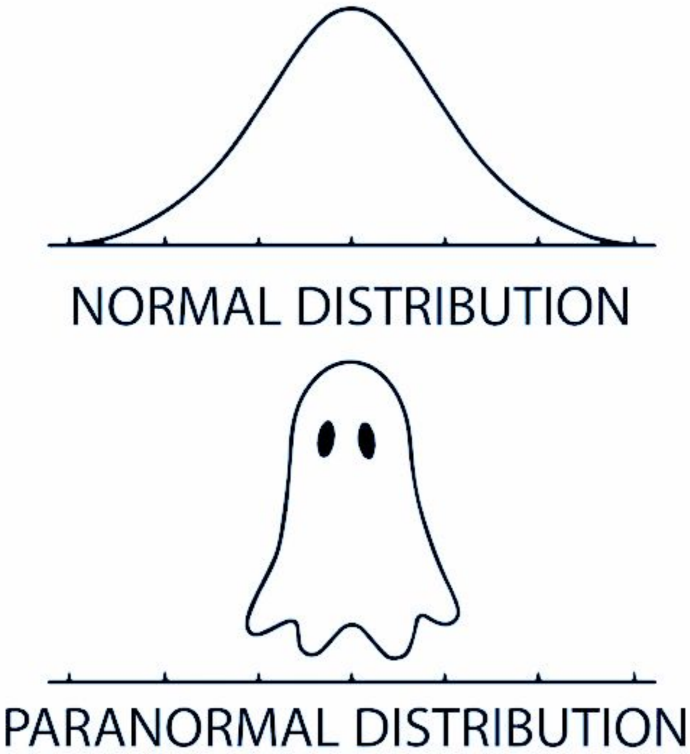
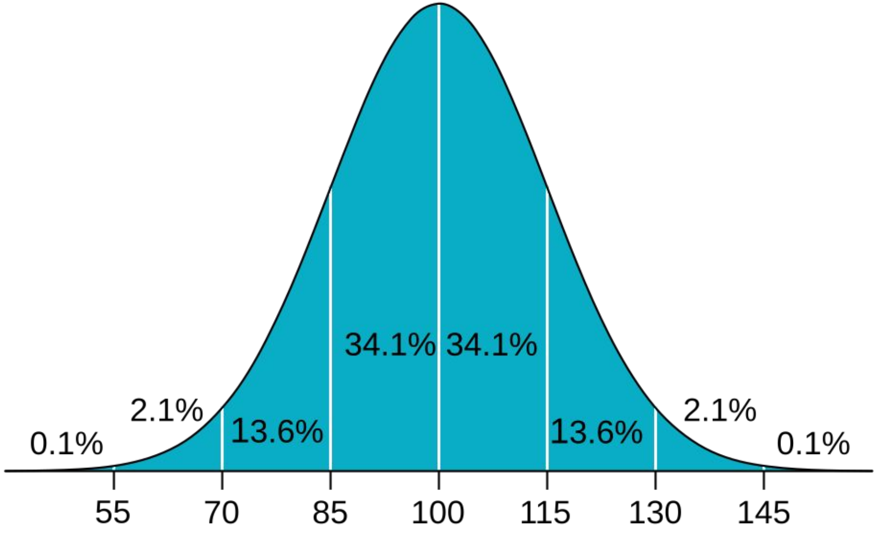
Normality Tests in Python



Normal Distributions are needed for hypothesis tests

Student's T-Test and Analysis of Variance (ANOVA) require that the distributions involved are normal.

How do we find out if a distribution is normal?



IQ distribution – a perfect example of a n.d.

(*numiqo*, 2025)
(*wikimedia*, 2013)

There are 3 popular normality tests

Shapiro-Wilk
$n \leq 2000$
contrasts X with n.d.

p-value > 0.05: may be n.d.
p-value ≤ 0.05: not n.d.

Kolmogorov-Smirnov (K-S)
$n > 2000$
compares EDF with CDF of n.d.

p-value > 0.05: could be n.d.
p-value ≤ 0.05: not n.d.
**sensitive to outliers*

Anderson-Darling
$n \in \mathbb{N}$
detects non-normality in extremes

α (or CV) < A^2 (or AD): could be n.d.
$\alpha \geq A^2$: not n.d.

With every test, we assume n.d., but judge by the results

(*numiqo*, 2025)
(*GeeksforGeeks*, 2025)

The QR to this presentation will be at the end

Shapiro-Wilk

$$W = \frac{\left(\sum_{i=1}^{n} a_i\, x_{(i)}\right)^2}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$$

$x_{(i)}$	i-th order statistic (sorted data)
a_i	coefficients derived from expected values of normal order statistics
$\bar{x}$	sample mean
$W \in (0,\, 1]$	ratio of best linear estimate of σ to the ordinary estimate

Kolmogorov-Smirnov

$$D = \sup_{x} \left| F_n(x) - F_0(x) \right|$$

$F_n(x)$	empirical CDF of the sample
$F_0(x)$	theoretical CDF of the reference distribution
$\sup_x$	supremum — largest absolute gap across all x
D	maximum vertical distance between the two CDFs

Anderson-Darling

$$A^2 = -n - \frac{1}{n} \sum_{i=1}^{n} (2i-1) \left[\ln F(x_{(i)}) + \ln\left(1 - F(x_{(n+1-i)})\right) \right]$$

n	sample size
$x_{(i)}$	i-th order statistic
$F(\cdot)$	CDF of the reference distribution evaluated at each data point
$\frac{2i-1}{n}$	weight term — amplifies deviations in the tails
A^2	weighted sum of squared CDF deviations; compared to a critical value

(*minitab*)

The tests are one-liners in Python

Shapiro-Wilk

```
stat, p = shapiro(data)
```

Kolmogorov-Smirnov

```
stat, p = kstest(data, "norm", args=(data.mean(), data.std(ddof=1)))
```

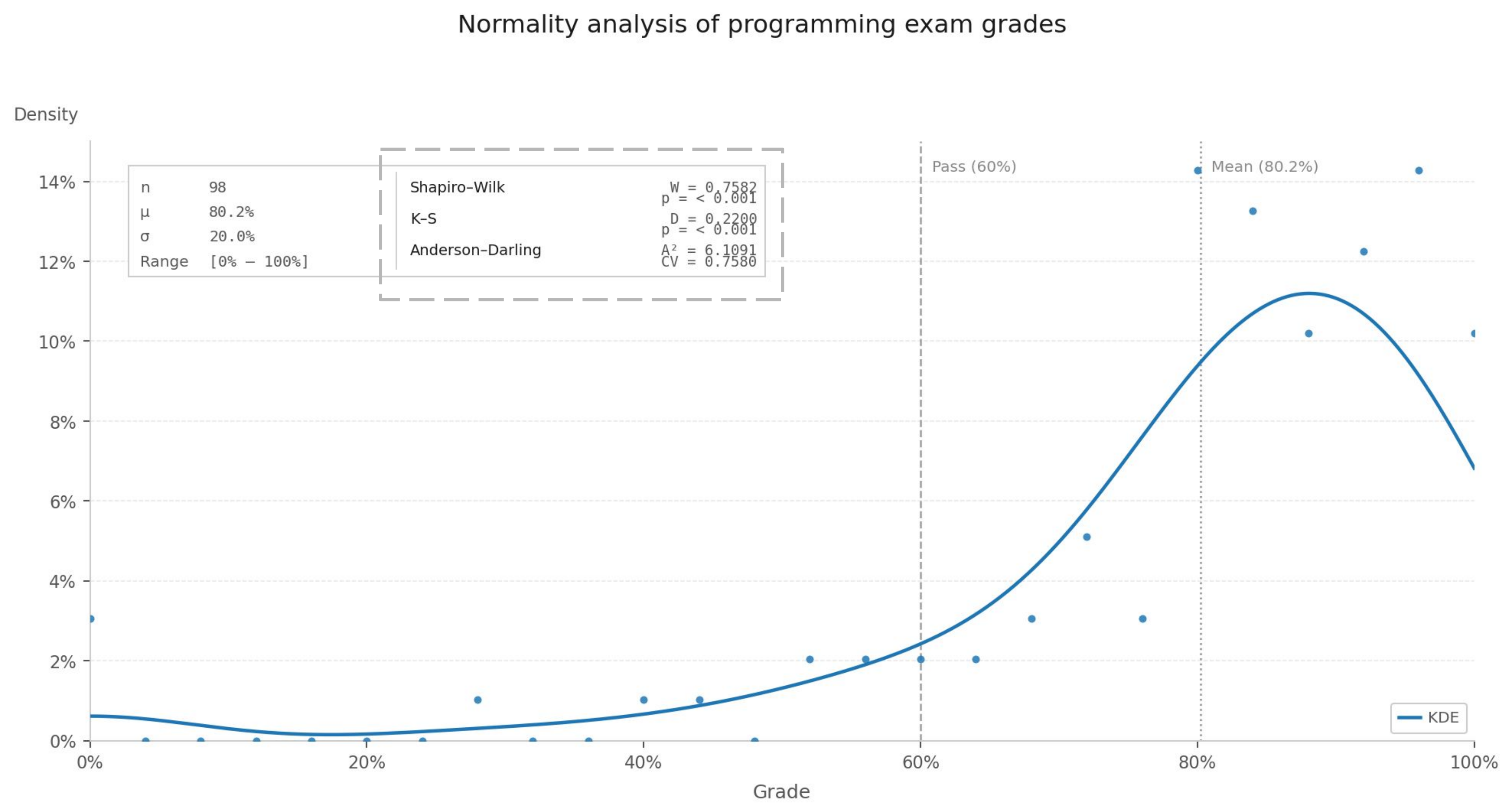
Anderson-Darling

```
result = anderson(data, dist="norm")
```

Let's run these on exam scores

[Sample Code]

(*scipy*)


Normality analysis of programming exam grades
Density
n 98
μ 80.2%
σ 20.0%
Range [0% – 100%]
Shapiro–Wilk
W = 0.7582
p = < 0.001
K–S
D = 0.2200
p = < 0.001
Anderson–Darling
A² = 6.1091
CV = 0.7580
Pass (60%)
Mean (80.2%)
KDE
14%
12%
10%
8%
6%
4%
2%
0%
0%
20%
40%
60%
80%
100%
Grade

Shapiro–Wilk	$W = 0.7582$ $p = < 0.001$
K–S	$D = 0.2200$ $p = < 0.001$
Anderson–Darling	$A^2 = 6.1091$ $CV = 0.7580$

Reject H_0

References

GeeksforGeeks. (2025, July 23). *Normality test*. GeeksforGeeks. https://www.geeksforgeeks.org/data-science/normality-test/

Statistical functions (scipy.stats) — SciPy v1.17.0 Manual. (n.d.). https://docs.scipy.org/doc/scipy/reference/stats.html

Test of normality: Assessing data distribution. (2025, September 1). https://numiqo.com/tutorial/test-of-normality

Methods and formulas for Normality Test - Minitab. (n.d.). (C) Minitab, LLC. All Rights Reserved. 2026. https://support.minitab.com/en-us/minitab/help-and-how-to/statistics/basic-statistics/how-to/normality-test/methods-and-formulas/methods-and-formulas/

This Presentation